

The empirical recurrence for OEIS A312928, proved from an exact model

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Abstract

OEIS A312928 carries an empirical recurrence of order 6 contributed by Chai Wah Wu. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by a certified distance function on the tiling, counted by Ehrhart, from which a provably satisfies a MONIC linear recurrence of order $S = 10$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A312928 is “Coordination sequence Gal.3.16.2 where Gal.u.t.v denotes the coordination sequence for a vertex of type v in tiling number t in the Galebach list of u-uniform tilings.”. It has offset 0 and begins

$$1, 4, 9, 13, 18, 22, 26, 31, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

Conjectures from _Chai Wah Wu_, Feb 14 2020: (Start) $a(n) = a(n-1) + a(n-5) - a(n-6)$ for $n > 6$.

As of the “Last modified” line on the live entry (Oct 21 2023 15:10:45, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The tiling, the lattice, and the distance

There is no array here and no transfer matrix. The entry names a vertex of one of Brian Galebach's k -uniform tilings, and that tiling is rebuilt exactly: the expanded notation fixes, for every vertex type, the angles between its edges and the type and frame of the vertex at the far end of each. All angles are multiples of 15° , so every vertex position is a $\mathbb{Z}$ -combination of 24th roots of unity and lives in $\mathbb{Z}[\zeta_{24}] = \mathbb{Z}^8$ modulo $x^8 - x^4 + 1$. Vertex identity is therefore decided by integer equality and not by a tolerance.

Let L be the group of translations of the plane carrying the tiling to itself. Two independent generators are found and each is VERIFIED to be a symmetry: for every vertex w of an inner patch, $w + t$ is a vertex of the same type and the neighbours of $w + t$ are exactly the neighbours of w translated. (Preserving a vertex's type and its set of edge directions is NOT sufficient, because a vertex figure with a rotational symmetry has several frames with the same direction

set.) The vertices fall into 5 orbits under L , and a vertex is written (c, m) with c its orbit and $m \in \mathbb{Z}^2$ its coordinates in a basis of L .

Write $D_c(m)$ for the graph distance from the entry's own vertex to (c, m) . The claim proved below is that on each orbit D_c is a maximum of finitely many affine functions,

$$D_c(m) = \max_i (A_{c,i}m_1 + B_{c,i}m_2 + C_{c,i}).$$

3 Why that is the distance, and not merely a fit

The affine pieces are obtained from a breadth-first patch, but nothing below rests on that. A function D with $D(\text{base}) = 0$ satisfying

- (b) $D_{c'}(m + d) \leq D_c(m) + 1$ for every edge (c, c', d) of the quotient and every m ;
- (c) every vertex other than the base has an edge attaining $D_c(m) = D_{c'}(m + d) + 1$;

IS the distance. (b) gives $D \leq d$ by induction along a shortest path, and (c) gives $D \geq d$ by induction on D ; neither induction cares where D came from. So the fit is a candidate and these two conditions are the proof.

Both are decided exactly, and on REGIONS rather than on sampled points. The set where the i -th piece is the maximum is the polyhedron $P_{c,i} = \{m : \ell_{c,i}(m) - \ell_{c,j}(m) \geq 0 \text{ for all } j\}$, so (b) is an affine inequality on a polyhedron — true exactly when it holds at every vertex of that polyhedron and its gradient pairs non-negatively with every recession ray. Condition (c) is not an affine statement but a disjunction, and it says that $P_{c,i}$ is COVERED by the regions $S_k = \{m : \ell_{c,i}(m) - 1 - \ell''_{k,j}(m + d_k) \geq 0 \text{ for all } j\}$, one per neighbour k ; that is decided by subtracting the S_k from $P_{c,i}$ one at a time and asking whether any region is left. Every coefficient is an integer, and since only lattice points matter a violated constraint is written ≤ -1 , so no strict inequality occurs anywhere. A bounded box around the base is checked exhaustively and the four half-planes outside it are covered by the region arithmetic.

4 The bound

The ball of radius t meets orbit c in the lattice points of

$$P_c(t) = \{m \in \mathbb{Z}^2 : A_{c,i}m_1 + B_{c,i}m_2 \leq t - C_{c,i} \text{ for every } i\},$$

a polygon whose facet NORMALS are fixed and whose right-hand sides move linearly with t . Every candidate vertex of it is the meet of two facets and moves affinely in t , so whether a given vertex satisfies a given facet flips at most once, at one rational t ; past the largest such value — computed here, not assumed — the combinatorial type never changes again. By Ehrhart's theorem for such a family the lattice count is then a quasi-polynomial in t of degree 2, whose period divides the lcm of the 2×2 determinants of pairs of facet normals, since those are the only denominators a vertex can have. Write q for that lcm; here $S = 2q = 10$.

Summing over orbits, $|B(t)|$ is a degree-2 quasi-polynomial of period q past the onset, so $a(n) = |B(n)| - |B(n-1)|$ is quasi-LINEAR of period q there, and so is the residual of any fixed linear recurrence. A quasi-linear function of period q that vanishes at $2q$ consecutive indices past the onset vanishes at every later one, which is what makes the check below finite and complete rather than a sample. The counting itself is never done through the closed form: the lattice points are counted exactly at every radius, and the quasi-polynomial's only job is to supply q .

The annihilator is therefore $A(z) = (z^q - 1)^2$, of degree $S = 2q$ and with $A(0) = 1 \neq 0$, and it annihilates a from the onset on rather than everywhere: the indices below it are a genuine transient, exactly as $\dim S_0$ and $\dim S_2$ are for the cusp-form family. That costs nothing, because the residual below is computed term by term from the exact lattice count and the run of zeros the theorem uses lies entirely above the onset.

5 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j < S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^6 - \sum_i c_i z^{6-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a , so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a , hence every shift of a , hence any linear combination of shifts; that is r . A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. $\square$

Theorem 2.

$$a(n) = a(n-1) + a(n-5) - a(n-6)$$

for every $n > 6$.

Proof. The terms $a(0), \dots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 6+1$ onwards, and the run exceeds $S+6$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 6$ the recurrence fails on the entry's own published terms, so no larger range holds. $\square$

6 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 50 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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